Alexandros Tsigonias-Dimitriadis

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SUMMARY Researcher and engineer currently developing AI/ML-based applications at the European Central Bank. I have a strong background in Computer Science and Applied Mathematics and a proven track record of applying tools from a wide range of disciplines, including algorithms, optimization, machine learning, mechanism design, and economics in both practical and research contexts. My research has been published in leading conferences and journals across operations research, economics and computation, and algorithms.

EMPLOYMENT

European Central Bank, Germany
Graduate Programme Participant — Sept. 2024 - present
- Selected among 15 out of ≈1500 applicants for two one-year assignments and an independent research project.
- Currently working as an AI/ML Engineer on end-to-end development of LLM-driven applications used across Banking Supervision areas. Focusing on RAG internal tools and on a multi-label text classification problem.

Data Scientist — July 2023 - August 2024
- Started as a PhD trainee for one year, then transitioned into a full-time role before switching to the Graduate Programme.
- Evaluated the impact of the ECB's communication to general and specialized audiences and informed strategy using various data sources (e.g., social media, news, surveys), statistical methods, and custom metrics.
- Conducted research on central banking communication, exploring how the stances of Governing Council members are portrayed by media outlets, using tools from computational social science.

Universidad de Chile, Chile — Aug. 2022 - July 2023
Postdoctoral Researcher, Department of Industrial Engineering
Information and Computation in Market Design (ICMD) Group (*Host*: Prof. José Correa)
- Focused on resource allocation problems arising in online markets and platforms. Topics included fairness considerations in recommendation systems, optimization aspects of political apportionment, sequential decision-making, auctions, and pricing.
- Collaborated with industry researchers, including a research visit with the Algorithms & Optimization team of Google Research Zurich.

EDUCATION

Technical University of Munich, Germany — Sept. 2018 - July 2022
Ph.D. in Mathematics, TUM School of Computation, Information and Technology
Completed an interdisciplinary PhD program with specialization in Operations Research.
Thesis: Lotteries, Prophets, and Pandora's Box: A New Take on Classic Problems in Mechanism Design and Online Selection
Advisor: Prof. Andreas S. Schulz

National Technical University of Athens, Greece — Oct. 2012 - June 2018
Diploma (BSc & MEng) in Electrical and Computer Engineering
GPA: 8.88/10 (top 7%)
Major in Computer Science, Minor in Mathematics & Computer Networks
Thesis: Analysis of Opinion Formation Dynamics using Convex Optimization Techniques
Advisor: Prof. Dimitris Fotakis

Technical University of Munich, Germany Apr. - Sept. 2016
Erasmus semester, Department of Electrical and Computer Engineering

RESEARCH VISITS

Google Research, Switzerland May - July 2023
Host: Dr. Paul Dütting

Sapienza Università di Roma, Italy Mar. 2022
Host: Prof. Stefano Leonardi

Universidad de Chile, Chile Nov. - Dec. 2019
Host: Prof. José Correa

CODING & SOFTWARE

Programming: Python, R, SQL, C/C++, Matlab, JavaScript
Software and Platforms: Gurobi, Pytorch, Git, AWS, HuggingFace, Azure, LaTeX

AWARDS & SCHOLARSHIPS

COST Action STSM Grant 2022
Funding from the COST Action for research visit at the Sapienza University of Rome

DFG-AdONE start-up funding 2021-22
Funding from the German Research Foundation (DFG) for talented early career researchers

IGSSE PhD Scholarship 2018-21

RESEARCH
(Authors are listed alphabetically,
[C]: Conference,
[J]: Journal)

A Simple Framework for Fairness of Exposure in Online Content Recommendation
José Correa, Émile Naquin, Nicolás Stier-Moses, Alexandros Tsigonias-Dimitriadis, Victor Verdugo
[C]: *Submitted to* ICML'26 2026

New Combinatorial Insights for Monotone Apportionment
Javier Cembrano, José Correa, Ulrike Schmidt-Kraepelin, Alexandros Tsigonias-Dimitriadis, Victor Verdugo
[J]: **Mathematics of Operations Research** 2025
[C]: *2025 ACM-SIAM Symposium on Discrete Algorithms* **(SODA'25)** 2025

Prophet Secretary Against the Online Optimal
Paul Dütting, Evangelia Gergatsouli, Rojin Rezvan, Yifeng Teng, Alexandros Tsigonias-Dimitriadis
[J]: **Mathematics of Operations Research**, forthcoming 2025
[C]: *24th ACM Conference on Economics and Computation* **(EC'23)** 2023

Prophet Inequalities via the Expected Competitive Ratio
Tomer Ezra, Stefano Leonardi, Rebecca Reiffenhäuser, Matteo Russo, Alexandros Tsigonias-Dimitriadis
[J]: *ACM Transactions on Economics and Computation* **(ACM TEAC)** 2025
[C]: *19th Conference on Web and Internet Economics* **(WINE'23)** 2023

The Secretary Problem with Independent Sampling
José Correa, Andrés Cristi, Laurent Feuilloley, Tim Oosterwijk, Alexandros Tsigonias-Dimitriadis
[J]: **Management Science** 2024
[C]: *2021 ACM-SIAM Symposium on Discrete Algorithms* **(SODA'21)** 2021

Preference Elicitation in Combinatorial Auctions: An Improved Approach using Reinforcement Learning

Eleni Batziou, Giorgos Bouritsas, Alexandros Tsigonias-Dimitriadis
Manuscript, available upon request 2023

Robust Revenue Maximization Under Minimal Statistical Information
Yiannis Giannakopoulos, Diogo Poças, Alexandros Tsigonias-Dimitriadis
[J]: *ACM Transactions on Economics and Computation* (ACM TEAC) 2022
[C]: *16th Conference on Web and Internet Economics* **(WINE'20)** 2020

PROFESSIONAL SERVICE

Conference & Journal reviewing: NeurIPS ('23, '25), ICML ('23, '26), Math. Programming
Conference subreviewing: COLT'23, IPCO'23, SODA'23, SAGT'22, EC'22
Volunteer, STOC 2022
Organizer of the AdONE research group retreat 2020

TEACHING & SUPERVISION

European Central Bank & Barcelona School of Economics
Co-supervisor, Master thesis in Data Science Spring 2024

Technical University of Munich, Germany
Teaching Assistant, Special Topics in Algorithmic Game Theory Spring 2019
Co-supervisor of two Master theses in the Department of Mathematics

National Technical University of Athens, Greece
Teaching Assistant, Algorithms and Complexity Fall 2017

Private Tutor, Athens, Greece 2013 - 2016
Mathematics and Physics for high school students

LANGUAGES

English (fluent), Greek (native), Spanish (fluent), German (advanced), French (basic)